

COURSE TITLE : ANALYTICAL INSTRUMENTATION
COURSE CODE : 6212
COURSE CATEGORY : A
PERIODS/WEEK : 5
PERIODS/SEMESTER: 75
CREDITS : 5

TIME SCHEDULE

Module	Topics	Period
1	Basics of analysis and absorption spectrometers	18
2	Emission Spectrometers	19
3	pH and Chromatography	19
4	Industrial Analyzers	19
TOTAL		75

COURSE OUTCOME

Module	G.O.	On completion of the study of this module the student will be able
1	1	To understand the basics of analysis
	2	To comprehend the working of absorption spectrometers
	3	To comprehend the working of Infra Red spectrometers
2	1	To understand the working of emission spectrometers
3	1	To understand pH measurement
	2	To understand the working of chromatograph
4	1	To comprehend the working of different industrial analyzers

On completion of the study the student will be able

MODULE I BASICS OF ANALYSIS AND ABSORPTION SPECTROMETERS

1.1.0 To understand the basics of analysis

- 1.1.1 To explain the various regions of electromagnetic spectrum
- 1.1.2 To describe the types of spectrum (absorption and emission)
- 1.1.3 To illustrate the fundamental laws of photometry.
- 1.1.4 To explain the basic components of photometer

1.2.0 To comprehend the working of absorption spectrometers

- 1.2.1 To differentiate filter photometers and spectrophotometers
- 1.2.2 To explain the principle, construction and working of single beam filter photometer
- 1.2.3 To illustrate the principle, construction and working of double beam filter photometer
- 1.2.4 To explain the principle, construction and working of single beam spectrophotometer
- 1.2.5 To describe the principle, construction and working of double beam spectrophotometer

1.3.0 To comprehend the working of Infrared spectrophotometer

- 1.3.1 To describe the basic components of infrared spectrophotometer.
- 1.3.2 To explain the working of I.R spectrophotometer

MODULE II EMISSION SPECTROMETERS

2.1.0 To understand the working of emission spectrometers

- 2.1.1 To explain the working of flame photometer
- 2.1.2 To describe the method of analysis by using flame photometer
- 2.1.3 To illustrate the principle of mass spectrometer
- 2.1.4 To explain the construction & working of magnetic deflection mass spectrometer
- 2.1.5 To explain the construction & working of time of flight mass spectrometer.
- 2.1.6 To describe the principle, construction & working of N.M.R spectrometer
- 2.1.7 To State the principle of Raman Effect
- 2.1.8 To explain the construction & working of Raman spectrophotometer

MODULE III PH AND CHROMATOGRAPHY

3.1.0 To understand pH measurement and control

- 3.1.1 To define pH and pH scale
- 3.1.2 To describe the importance of Buffer solution
- 3.1.3 To explain the construction and working of Hydrogen electrode
- 3.1.4 To illustrate the construction and working of glass electrode
- 3.1.5 To List the sources of errors in glass electrode
- 3.1.6 To explain the methods of error compensation in glass electrode
- 3.1.7 To describe the construction and working of calomel electrode
- 3.1.8 To explain the construction and working of combined pH electrode.
- 3.1.9 To Explain industrial electrode assembly dip and flow type
- 3.1.10 To describe the working of digital pH meter
- 3.1.11 To Explain pH control in effluent treatment

3.2.0 To understand Chromatography

- 3.2.1 To explain the importance of Chromatograph
- 3.2.2 To describe the Classification of chromatography
- 3.2.3 To Specify the principle of gas chromatograph
- 3.2.4 To explain the construction and working of gas chromatograph
- 3.2.5 To specify the principle of liquid chromatograph
- 3.2.6 To explain the construction and working of liquid chromatograph
- 3.2.7 To describe the construction and working of paper chromatograph

MODULE IV INDUSTRIAL ANALYZERS

4.1.0 To comprehend the working of Industrial Analyzers

- 4.1.1 To state the principle of Thermal conductivity analyzer
- 4.1.2 To explain the -construction and operation of Thermal conductivity gas analyzer
- 4.1.3 To state the principle of Paramagnetic analyzer
- 4.1.4 To explain the construction and operation of magnetic force type and wind type Paramagnetic oxygen analyzers
- 4.1.5 To describe the construction and operation of Zirconia oxygen analyzer
- 4.1.6 To state the principle of Infrared analyzers
- 4.1.7 To Classify Infrared analyzers
- 4.1.8 To explain the construction and operation of positive filter type IR analyzer
- 4.1.9 To illustrate the construction and operation of negative filter type IR analyzer
- 4.1.10 To State the principle of Electrical conductivity analyzer
- 4.1.11 To describe the construction and operation of Electrical conductivity analyzer
- 4.1.12 To explain Measurement circuit used in Electrical conductivity analyzer

CONTENT DETAILS

MODULE I

U.V and Visible photometers – Electromagnetic spectrum – types of spectrum - Absorption spectrometers -fundamental laws of absorption photometry – components of photometer -filter photometer – single beam and double beam -spectrophotometers – single beam and double beam - I.R. spectrophotometers.

MODULE II

Emission Spectrometers - flame photometer - Mass spectrophotometer– construction and working – Magnetic deflection mass spectrometer -Time of flight mass spectrometer –Nuclear Magnetic Resonance spectrometer– Raman effect -Raman spectrophotometer.

MODULE III

pH Measurement - Introduction – Sorensen's scale – Buffer solution – electrodes for pH measurements, measuring electrodes-Hydrogen electrode, glass electrode, errors and its compensation in glass electrode- calomel electrode-combined pH electrode -Industrial electrode assemblies-DIP and Flow type Digital pH meter- pH control in effluent treatment- Chromatography -Classification - Gas chromatography–Liquid chromatography-Paper chromatography .

MODULE IV

Industrial Analyzers -Thermal conductivity gas analyzer— Paramagnetic oxygen analyzer – magnetic force type and wind type - Zirconia oxygen analyzer - Infrared analyzers – positive and negative filter type IR analyzer- Electrical conductivity analyzer .

REFERENCES:

1. Williard, Dean Merrit & Settle , Instrumental methods of analysis- CBS publishers
2. R.S.Khandpur , Handbook of Analytical Instruments- TMH
3. Chatwal – Anand , Instrumental methods of Chemical Analysis - Himalaya Publishing House
4. R K Jain- Mechanical and Industrial Measurements- Khanna Publishers
5. B.L.Theraja ,Engineering Physics – S Chand